

UNIT I - COMPUTATIONAL THINKING AND PROBLEM SOLVING

Fundamentals of Computing, Identification of Computational problems, Algorithms, building blocks of algorithms (statements, state, control flow, functions), notation (pseudo code, flow chart, programming language), algorithmic problem solving, simple strategies for developing algorithms (iteration, recursion). Illustrative problems: find minimum in a list, insert a card in a list of sorted cards, Guess an integer number in a range, Towers of Hanoi.

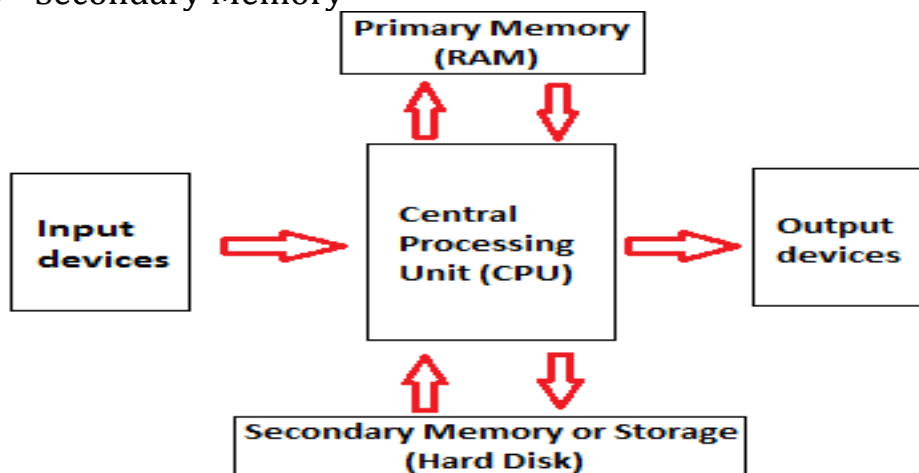
1.FUNDAMENTALS OF COMPUTING

- A **computer** is an electronic device that accepts data from the user, processes it, produces results, displays them to the users, and stores the results for future usage.
- **Data** is a collection of unorganized facts & figures and does not provide any further information regarding patterns, context, etc. Hence data means "unstructured facts and figures".
- **Information** is a structured data i.e. organized meaningful and processed data. To process the data and convert into information, a computer is used

COMPUTER PARTS OR COMPONENTS

Computer components are the essential building parts of developing a functional computer system. The components that make up a computer are called computer components. There are 5 main computer components that are given below

- Input Devices
- CPU
- Output Devices
- Primary Memory
- Secondary Memory



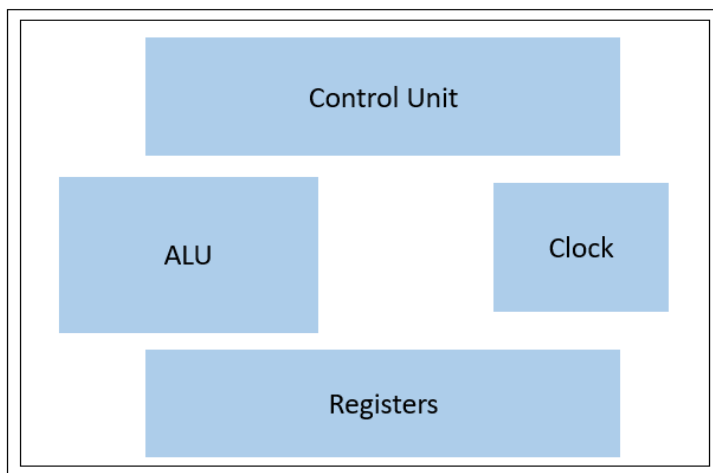
Input devices:

A computer system's input devices are important because they allow users to enter commands and data. Keyboards, mice, scanners, and microphones are numerous examples of input devices.

CPU:

A computer machine's "brain" is its central processing unit (CPU). It executes the calculations and commands required for functioning of the computer device. The CPU comprises some components: the control unit, the arithmetic logic unit (ALU), and registers.

CPU



- The CPU's control unit is in charge of reading and decoding instructions from memory. The right part of the CPU receives and executes these instructions from the control unit.
- The ALU operates addition, subtraction, comparison calculations, and other logical and mathematical processes.
- Registers are compact, high-speed data and instruction storage spaces within the CPU.
- The clock speed, measured in GHz (gigahertz), controls what number of commands the integral processing unit can process in a second.

Primary Memory:

The CPU has direct access to primary memory, sometimes referred to as random access memory (RAM). The data and instructions that are currently being processed are kept in primary memory. The data and instructions are accessed by the CPU from primary memory when a computer programme is running. The information is removed from primary memory once the programme is completed.

Primary memory is classified into two types: random access memory (RAM) and read-only memory (ROM).

- **RAM** is the most common form of primary memory and is used to store data and instructions that the CPU wishes to access frequently. RAM is volatile, which means that its contents are lost when the computer is turned off. But RAM can be effortlessly and quickly written to and read from, making it a really perfect storage medium for temporary data and instructions.
- **ROM** is a form of memory this is used to store data and instructions that don't change. ROM is non-volatile, which means that its contents aren't lost while the computer is turned off. ROM is used to keep firmware and the laptop's basic input/output machine (BIOS), that are required for the computer to boot up and function well.

Secondary Memory:

Secondary memory, also called auxiliary storage, is a type of computer memory that is used to store data and programs that aren't currently being utilized by the CPU. In contrast to primary memory, secondary memory is non-volatile, which means that its contents are not lost when the computer is turned off.

There are several types of secondary memory devices, such as hard disk drives (HDD), solid-state drives (SSD), optical disks (including CDs and DVDs), and USB flash drives.

Output Devices:

Output devices are hardware components of a computer system that are used to show or send data from the pc to the user or any other device. They enable customers to view and engage with the information and applications the computer is processing. Speakers, projectors, printers, and monitors are a few examples of output devices.

FUNCTIONS OF COMPUTERS

The operations of computer components are given below:

Computer components collaborate to carry out the numerous tasks necessary for a computer system to run. The following are some of the major operations carried out by computer components:

1. **Inputting:** It is the process of entering raw data, instructions and information into the computer. Keyboards, mice, and scanners are used as input devices to help with the process. These tools are used to enter information and instructions into a computer system. Data is transferred to the CPU for processing after inputting by an input device.
2. **Processing:** It is the process of converting the raw data into useful information. This process is performed by the CPU of the computer. It takes the raw data from storage, processes it and then sends back the processed data to storage. The CPU performs arithmetic computations, logical operations, and data transport processes.

3. **Storing:** The computer has primary memory and secondary storage to store data and instructions. It stores the data before sending it to CPU for processing and also stores the processed data before displaying it as output. The primary memory, sometimes called RAM, is where the CPU processes the data and instructions. Hard disc drives and solid-state drives, which serve as secondary memory, offer long-term storage for data and programmes that are not currently used.
4. **Outputting:** It is the process of presenting the processed data through output devices like monitor, printer and speakers. These devices display or produce the results of the processing performed by the CPU. The results are sent to an output device for display or printing after the CPU has finished processing the data and instructions.
5. **Controlling:** This operation is performed by the control unit that is part of CPU. The control unit ensures that all basic operations are executed in a right manner and sequence. The main circuit board connects all the parts of the computer. It also regulates the data flow between them, ensuring they function properly and connecting them all.

2.IDENTIFICATION OF COMPUTATIONAL PROBLEMS

A computational problem is a problem that can be solved step-by-step with a computer. These problems usually have a well-defined input, constraints, and conditions that the output must satisfy. Here are some types of computational problems:

- A **decision problem** is one where the answer is yes or no. For instance, "given a number n , is n even?" is a decision problem. Some decision problems take more steps to solve than others. For instance, "given a number n , is n prime?" takes more steps than just checking the parity of a number.
- A **search problem** is one where the solution consists of one or more values that satisfy a given condition. For instance, we may want to compute a path from one geographical location to another on a map.
- A **counting problem** is one where the answer is the number of solutions to a search problem.
- An **optimization problem** is one where the solution is the "best" possible solution, where the "best" can be defined in a different way. For instance, we may want to compute the fastest route from one location to another.

3.PROBLEM SOLVING

Problem solving is the systematic approach to define the problem and creating number of solutions.

The problem solving process starts with the problem specifications and ends with a

Correct program.

3.1 PROBLEM SOLVING TECHNIQUES

Problem solving technique is a set of techniques that helps in providing logic for solving a problem.

Problem Solving Techniques:

Problem solving can be expressed in the form of

1. Algorithms.
2. Flowcharts.
3. Pseudo codes.
4. programs

1.2. ALGORITHM

It is defined as a sequence of instructions that describe a method for solving a problem. In other words it is a step by step procedure for solving a problem.

Properties of Algorithms

- ❖ Should be written in simple English
- ❖ Each and every instruction should be precise and unambiguous.
- ❖ Instructions in an algorithm should not be repeated infinitely.
- ❖ Algorithm should conclude after a finite number of steps.
- ❖ Should have an end point
- ❖ Derived results should be obtained only after the algorithm terminates.

Qualities of a good algorithm

The following are the primary factors that are often used to judge the quality of the algorithms.

Time – To execute a program, the computer system takes some amount of time. The lesser is the time required, the better is the algorithm.

Memory – To execute a program, computer system takes some amount of memory space. The lesser is the memory required, the better is the algorithm.

Accuracy – Multiple algorithms may provide suitable or correct solutions to a given problem, some of these may provide more accurate results than others, and such algorithms may be suitable.

Example:

Example

Write an algorithm to print „Good Morning”

Step 1: Start

Step 2: Print “Good Morning”

Step 3: Stop

2. BUILDING BLOCKS OF ALGORITHMS (statements, state, control flow, functions)

Algorithms can be constructed from basic building blocks namely, sequence, selection and iteration.

2.1. Statements:

Statement is a single action in a computer.

In a computer statements might include some of the following actions

- input data-information given to the program
- process data-perform operation on a given input

➤ output data-processed result

2.2. State:

Transition from one process to another process under specified condition with in a time is called state.

2.3. Control flow:

The process of executing the individual statements in a given order is called control flow.

The control can be executed in three ways

1. sequence
2. selection
3. iteration

Sequence:

All the instructions are executed one after another is called sequence execution.

Example:

Add two numbers:

Step 1: Start

Step 2: get a,b

Step 3: calculate $c=a+b$

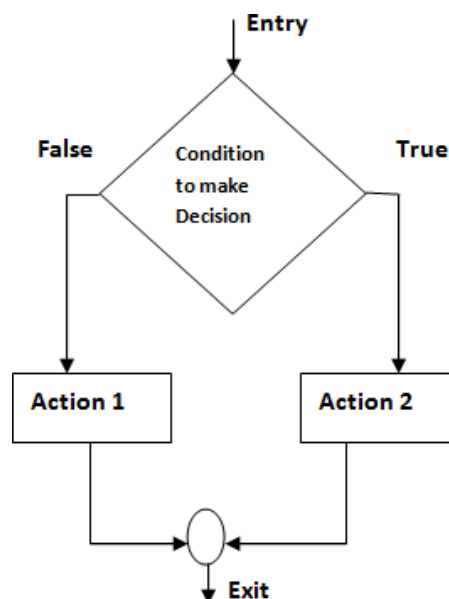
Step 4: Display c

Step 5: Stop

Selection:

A selection statement causes the program control to be transferred to a specific part of the program based upon the condition.

If the conditional test is true, one part of the program will be executed, otherwise it will execute the other part of the program.



Example

Write an algorithm to check whether he is eligible to vote?

Step 1: Start

Step 2: Get age

Step 3: if age $\geq$ 18 print "Eligible to vote"

Step 4: else print "Not eligible to vote"

Step 6: Stop

Iteration:

In some programs, certain set of statements are executed again and again based upon conditional test. i.e. executed more than one time. This type of execution is called looping or iteration.

Example

Write an algorithm to print all natural numbers up to n

Step 1: Start

Step 2: get n value.

Step 3: initialize i=1

Step 4: if (i $\leq$ n) go to step 5 else go to step 7

Step 5: Print i value and increment i value by 1

Step 6: go to step 4

Step 7: Stop

2.4. Functions:

- ❖ Function is a sub program which consists of block of code(set of instructions) that performs a particular task.
- ❖ For complex problems, the problem is been divided into smaller and simpler tasks during algorithm design.

Benefits of Using Functions

- ❖ Reduction in line of code
- ❖ code reuse
- ❖ Better readability
- ❖ Information hiding
- ❖ Easy to debug and test
- ❖ Improved maintainability

Example:

Algorithm for addition of two numbers using function

Main function()

Step 1: Start

Step 2: Call the function add()

Step 3: Stop

sub function add()

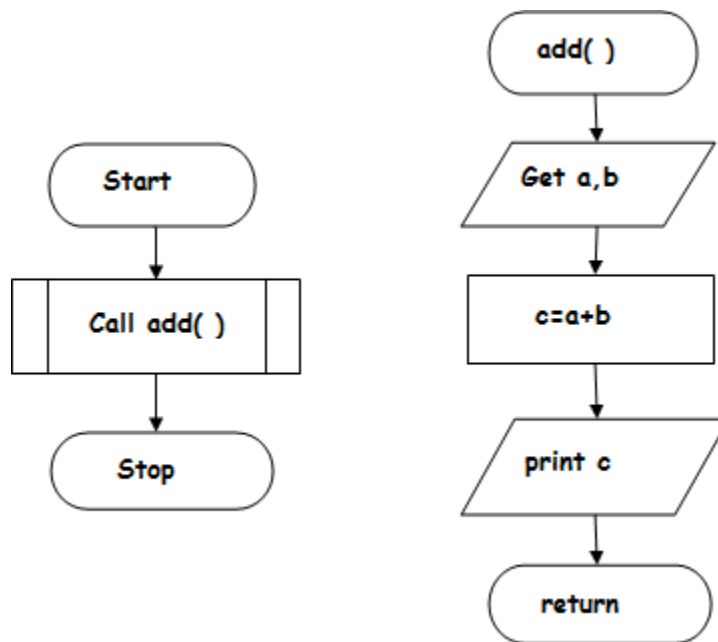
Step 1: Function start

Step 2: Get a, b Values

Step 3: add c=a+b

Step 4: Print c

Step 5: Return



3. NOTATIONS

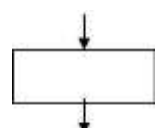
3.1. FLOW CHART

Flow chart is defined as graphical representation of the logic for problem solving. The purpose of flowchart is making the logic of the program clear in a visual representation.

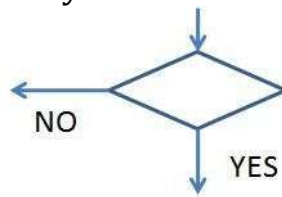
Symbol	Symbol Name	Description
	Flow Lines	Used to connect symbols
	Terminal	Used to start, pause or halt in the program logic
	Input/output	Represents the information entering or leaving the system
	Processing	Represents arithmetic and logical instructions
	Decision	Represents a decision to be made
	Connector	Used to Join different flow lines
	Sub function	used to call function

Rules for drawing a flowchart

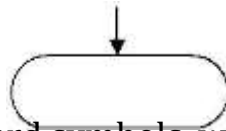
1. The flowchart should be clear, neat and easy to follow.
2. The flowchart must have a logical start and finish.
3. Only one flow line should come out from a process symbol.



4. Only one flow line should enter a decision symbol. However, two or three flow lines may leave the decision symbol.



5. Only one flow line is used with a terminal symbol.



6. Within standard symbols, write briefly and precisely.
7. Intersection of flow lines should be avoided.

Advantages of flowchart:

1. **Communication:** - Flowcharts are better way of communicating the logic of a system to all concerned.
2. **Effective analysis:** - With the help of flowchart, problem can be analyzed in more effective way.
3. **Proper documentation:** - Program flowcharts serve as a good program documentation, which is needed for various purposes.
4. **Efficient Coding:** - The flowcharts act as a guide or blueprint during the systems analysis and program development phase.
5. **Proper Debugging:** - The flowchart helps in debugging process.
6. **Efficient Program Maintenance:** - The maintenance of operating program becomes easy with the help of flowchart. It helps the programmer to put efforts more efficiently on that part.

Disadvantages of flow chart:

1. **Complex logic:** - Sometimes, the program logic is quite complicated. In that case, flowchart becomes complex and clumsy.
2. **Alterations and Modifications:** - If alterations are required the flowchart may require re-drawing completely.
3. **Reproduction:** - As the flowchart symbols cannot be typed, reproduction of flowchart becomes a problem.
4. **Cost:** For large application the time and cost of flowchart drawing becomes costly.

3.2.PSEUDO CODE:

- ☑ Pseudo code consists of short, readable and formally styled English languages used for explain an algorithm.
- ☑ It does not include details like variable declaration, subroutines.
- ☑ It is easier to understand for the programmer or non programmer to understand the general working of the program, because it is not based on any programming language.
- ☑ It gives us the sketch of the program before actual coding.

- ☒ It is not a machine readable
- ☒ Pseudo code can't be compiled and executed.
- ☒ There is no standard syntax for pseudo code.

Guidelines for writing pseudo code:

- ☒ Write one statement per line
- ☒ Capitalize initial keyword
- ☒ Indent to hierarchy
- ☒ End multiline structure
- ☒ Keep statements language independent

Common keywords used in pseudocode

The following gives common keywords used in pseudocodes.

1. **//**: This keyword used to represent a comment.
2. **BEGIN,END**: Begin is the first statement and end is the last statement.
3. **INPUT, GET, READ**: The keyword is used to inputting data.
4. **COMPUTE, CALCULATE**: used for calculation of the result of the given expression.
5. **ADD, SUBTRACT, INITIALIZE** used for addition, subtraction and initialization.
6. **OUTPUT, PRINT, DISPLAY**: It is used to display the output of the program.
7. **IF, ELSE, ENDIF**: used to make decision.
8. **WHILE, ENDWHILE**: used for iterative statements.
9. **FOR, ENDFOR**: Another iterative incremented/decremented tested automatically.

<u>Syntax for if else:</u>	<u>Example: Greates of two numbers</u>
IF (condition) THEN statement ... ELSE statement ... ENDIF	BEGIN READ a,b IF (a>b) THEN DISPLAY a is greater ELSE DISPLAY b is greater END IF END
<u>Syntax for For:</u>	<u>Example: Print n natural numbers</u>
FOR(<i>start-value</i> to <i>end-value</i>) DO <i>statement</i> ... ENDFOR	BEGIN GET n INITIALIZE i=1 FOR (i<=n) DO PRINT i i=i+1 ENDFOR END

<u>Syntax for While:</u>	<u>Example: Print n natural numbers</u>
WHILE (condition) DO statement ... ENDWHILE	BEGIN GET n INITIALIZE i=1 WHILE(i<=n) DO PRINT i i=i+1 ENDWHILE END

Advantages:

- ❖ Pseudo is independent of any language; it can be used by most programmers.
- ❖ It is easy to translate pseudo code into a programming language.
- ❖ It can be easily modified as compared to flowchart.
- ❖ Converting a pseudo code to programming language is very easy as compared with converting a flowchart to programming language.

Disadvantages:

- ❖ It does not provide visual representation of the program's logic.
- ❖ There are no accepted standards for writing pseudo codes.
- ❖ It cannot be compiled nor executed.
- ❖ For a beginner, It is more difficult to follow the logic or write pseudo code as compared to flowchart.

Example:

Addition of two numbers:

```

BEGIN
GET a,b
ADD c=a+b
PRINT c
END

```

Algorithm	Flowchart	Pseudo code
An algorithm is a sequence of instructions used to solve a problem	It is a graphical representation of algorithm	It is a language representation of algorithm.
User needs knowledge to write algorithm.	not need knowledge of program to draw or understand flowchart	Not need knowledge of program language to understand or write a pseudo code.

3.3.PROGRAMMING LANGUAGE

A programming language is a set of symbols and rules for instructing a computer to perform specific tasks. The programmers have to follow all the specified rules before writing program using programming language. The user has to communicate with the computer using language which it can understand.

Types of programming language

1. Machine language

2. Assembly language
3. High level language

Machine language:

The computer can understand only machine language which uses 0's and 1's. In machine language the different instructions are formed by taking different combinations of 0's and 1's.

Advantages:

Translation free:

Machine language is the only language which the computer understands. For executing any program written in any programming language, the conversion to machine language is necessary. The program written in machine language can be executed directly on computer. In this case any conversion process is not required.

High speed

The machine language program is translation free. Since the conversion time is saved, the execution of machine language program is extremely fast.

Disadvantage:

- It is hard to find errors in a program written in the machine language.
- Writing program in machine language is a time consuming process.

Machine dependent: According to architecture used, the computer differs from each other. So machine language differs from computer to computer. So a program developed for a particular type of computer may not run on other type of computer.

Assembly language:

- To overcome the issues in programming language and make the programming process easier, an assembly language is developed which is logically equivalent to machine language but it is easier for people to read, write and understand.
- Assembly language is symbolic representation of machine language. Assembly languages are symbolic programming language that uses symbolic notation to represent machine language instructions. They are called low level language because they are so closely related to the machines.

Ex: ADD a, b

Assembler:

Assembler is the program which translates assembly language instruction into a machine language.

Advantage:

- Easy to understand and use.
- It is easy to locate and correct errors.

Disadvantage

Machine dependent

The assembly language program which can be executed on the machine depends on the architecture of that computer.

Hard to learn

It is machine dependent, so the programmer should have the hardware knowledge to create applications using assembly language.

Less efficient

- Execution time of assembly language program is more than machine language program.
- Because assembler is needed to convert from assembly language to machine language.

High level language

High level language contains English words and symbols. The specified rules are to be followed while writing program in high level language. The interpreter or compilers are used for converting these programs in to machine readable form.

Translating high level language to machine language

The programs that translate high level language in to machine language are called interpreter or compiler.

Compiler:

A compiler is a program which translates the source code written in a high level language in to object code which is in machine language program. Compiler reads the whole program written in high level language and translates it to machine language. If any error is found it display error message on the screen.

Interpreter

Interpreter translates the high level language program in line by line manner. The interpreter translates a high level language statement in a source program to a machine code and executes it immediately before translating the next statement. When an error is found the execution of the program is halted and error message is displayed on the screen.

Advantages

Readability

High level language is closer to natural language so they are easier to learn and understand

Machine independent

High level language program have the advantage of being portable between machines.

Easy debugging

Easy to find and correct error in high level language

Disadvantages

Less efficient

The translation process increases the execution time of the program. Programs in high level language require more memory and take more execution time to execute.

They are divided into following categories:

1. Interpreted programming languages
2. Functional programming languages
3. Compiled programming languages
4. Procedural programming languages
5. Scripting programming language
6. Markup programming language
7. Concurrent programming language
8. Object oriented programming language

Interpreted programming languages:

An interpreted language is a programming language for which most of its implementation executes instructions directly, without previously compiling a program into machine language instructions. The interpreter executes the program directly translating each statement into a sequence of one or more subroutines already compiled into machine code.

Examples:

Pascal
Python

Functional programming language:

Functional programming language defines every computation as a mathematical evaluation. They focus on the programming languages are bound to mathematical calculations

Examples:

Clean
Haskell

Compiled Programming language:

A compiled programming is a programming language whose implementation are typically compilers and not interpreters.

It will produce a machine code from source code.

Examples:

C
C++
C#
JAVA

Procedural programming language:

Procedural (imperative) programming implies specifying the steps that the programs should take to reach to an intended state.

A procedure is a group of statements that can be referred through a procedure call. Procedures help in the reuse of code. Procedural programming makes the programs structured and easily traceable for program flow.

Examples:

Hyper talk
MATLAB

Scripting language:

Scripting language are programming languages that control an application. Scripts can execute independent of any other application. They are mostly embedded in the application that they control and are used to automate frequently executed tasks like communicating with external program.

Examples:

Apple script
VB script

Markup languages:

A markup language is an artificial language that uses annotations to text that define hoe the text is to be displayed.

Examples:

HTML
XML

Concurrent programming language:

Concurrent programming is a computer programming technique that provides for the execution of operation concurrently, either with in a single computer or across a number of systems.

Examples:

Joule
Limbo

Object oriented programming language:

Object oriented programming is a programming paradigm based on the concept of objects which may contain data in the form of procedures often known as methods.

Examples:

Lava
Moto

4.ALGORITHMIC PROBLEM SOLVING:

Algorithmic problem solving is solving problem that require the formulation of an algorithm for the solution.

Understanding the Problem

- ❖ It is the process of finding the input of the problem that the algorithm solves.
- ❖ It is very important to specify exactly the set of inputs the algorithm needs to handle.
- ❖ A correct algorithm is not one that works most of the time, but one that works correctly for *all* legitimate inputs.

Ascertaining the Capabilities of the Computational Device

- ❖ If the instructions are executed one after another, it is called sequential algorithm.
- ❖ If the instructions are executed concurrently, it is called parallel algorithm.

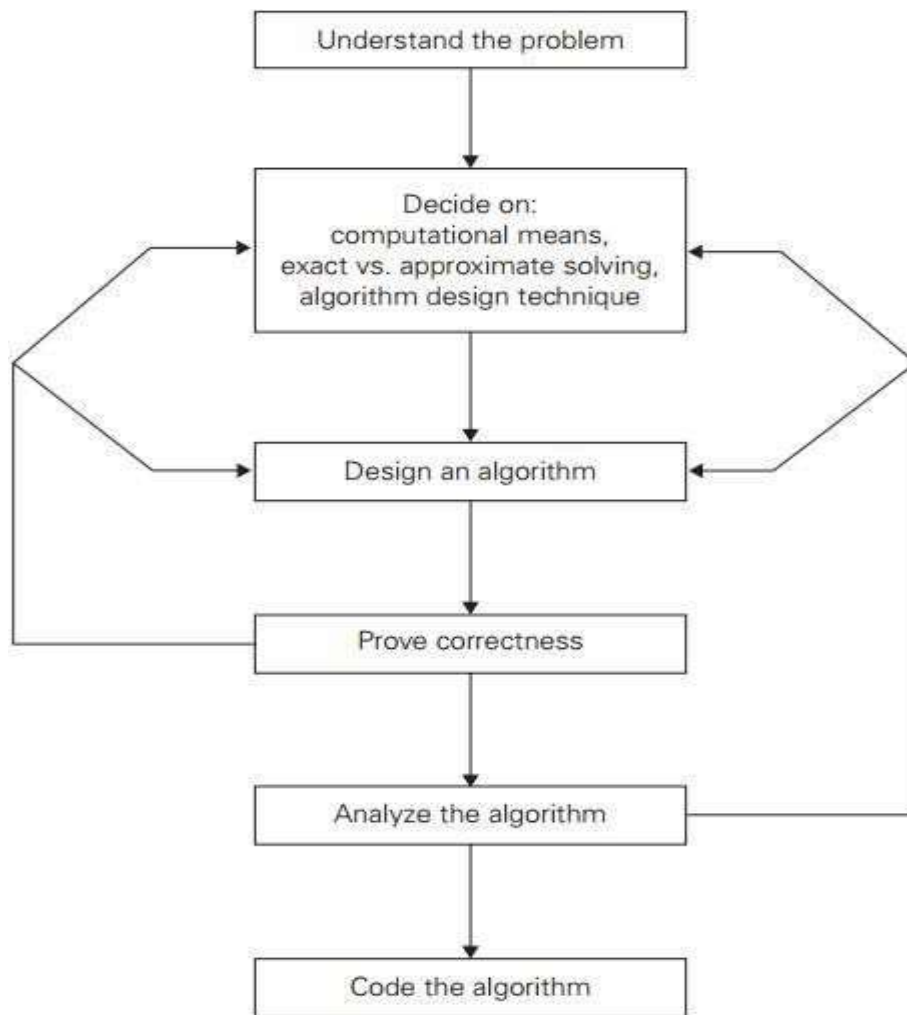


FIGURE 1.2 Algorithm design and analysis process.

Choosing between Exact and Approximate Problem Solving

- ❖ The next principal decision is to choose between solving the problem exactly or solving it approximately.
- ❖ Based on this, the algorithms are classified as exact *algorithm* and *approximation algorithm*.

Deciding a data structure:

- ❖ Data structure plays a vital role in designing and analysis the algorithms.
- ❖ Some of the algorithm design techniques also depend on the structuring data specifying a problem's instance
- ❖ Algorithm+ Data structure=programs.

Algorithm Design Techniques

- ❖ An *algorithm design technique* (or “strategy” or “paradigm”) is a general approach to solving problems algorithmically that is applicable to a variety of problems from different areas of computing.
- ❖ Learning these techniques is of utmost importance for the following reasons.
- ❖ First, they provide guidance for designing algorithms for new problems,
- ❖ Second, algorithms are the cornerstone of computer science

Methods of Specifying an Algorithm

- ❖ **Pseudocode** is a mixture of a natural language and programming language-like constructs. Pseudocode is usually more precise than natural language, and its usage often yields more succinct algorithm descriptions.
- ❖ In the earlier days of computing, the dominant vehicle for specifying algorithms was a **flowchart**, a method of expressing an algorithm by a collection of connected geometric shapes containing descriptions of the algorithm's steps.
- ❖ **Programming language** can be fed into an electronic computer directly. Instead, it needs to be converted into a computer program written in a particular computer language. We can look at such a program as yet another way of specifying the algorithm, although it is preferable to consider it as the algorithm's implementation.

Proving an Algorithm's Correctness

- ❖ Once an algorithm has been specified, you have to prove its **correctness**. That is, you have to prove that the algorithm yields a required result for every legitimate input in a finite amount of time.
- ❖ A common technique for proving correctness is to use mathematical induction because an algorithm's iterations provide a natural sequence of steps needed for such proofs.
- ❖ It might be worth mentioning that although tracing the algorithm's performance for a few specific inputs can be a very worthwhile activity, it cannot prove the algorithm's correctness conclusively. But in order to show that an algorithm is incorrect, you need just one instance of its input for which the algorithm fails.

Analysing an Algorithm

1. **Efficiency.**

Time efficiency, indicating how fast the algorithm runs,

Space efficiency, indicating how much extra memory it uses.

2. **simplicity.**

- ❖ An algorithm should be precisely defined and investigated with mathematical expressions.
- ❖ Simpler algorithms are easier to understand and easier to program.
- ❖ Simple algorithms usually contain fewer bugs.

Coding an Algorithm

- ❖ Most algorithms are destined to be ultimately implemented as computer programs. Programming an algorithm presents both a peril and an opportunity.
- ❖ A working program provides an additional opportunity in allowing an empirical analysis of the underlying algorithm. Such an analysis is based on timing the program on several inputs and then analysing the results obtained.

5.SIMPLE STRATEGIES FOR DEVELOPING ALGORITHMS:

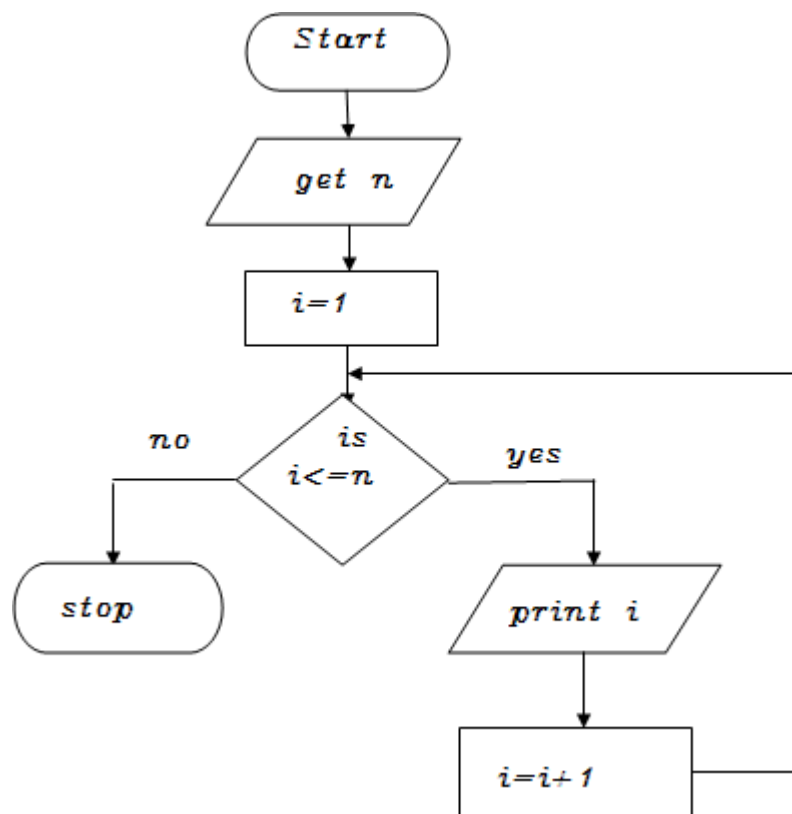
1. iterations
2. Recursions

5.1. Iterations:

A sequence of statements is executed until a specified condition is true is called iterations.

1. for loop
2. While loop

<u>Syntax for For:</u>	<u>Example: Print n natural numbers</u>
FOR(<i>start-value to end-value</i>) DO <i>statement</i> ... ENDFOR	BEGIN GET n INITIALIZE i=1 FOR (i<=n) DO PRINT i i=i+1 ENDFOR END
<u>Syntax for While:</u>	<u>Example: Print n natural numbers</u>
WHILE (condition) DO statement ... ENDWHILE	BEGIN GET n INITIALIZE i=1 WHILE(i<=n) DO PRINT i i=i+1 ENDWHILE END



5.2. Recursions:

- ❖ A function that calls itself is known as recursion.
- ❖ Recursion is a process by which a function calls itself repeatedly until some specified condition has been satisfied.

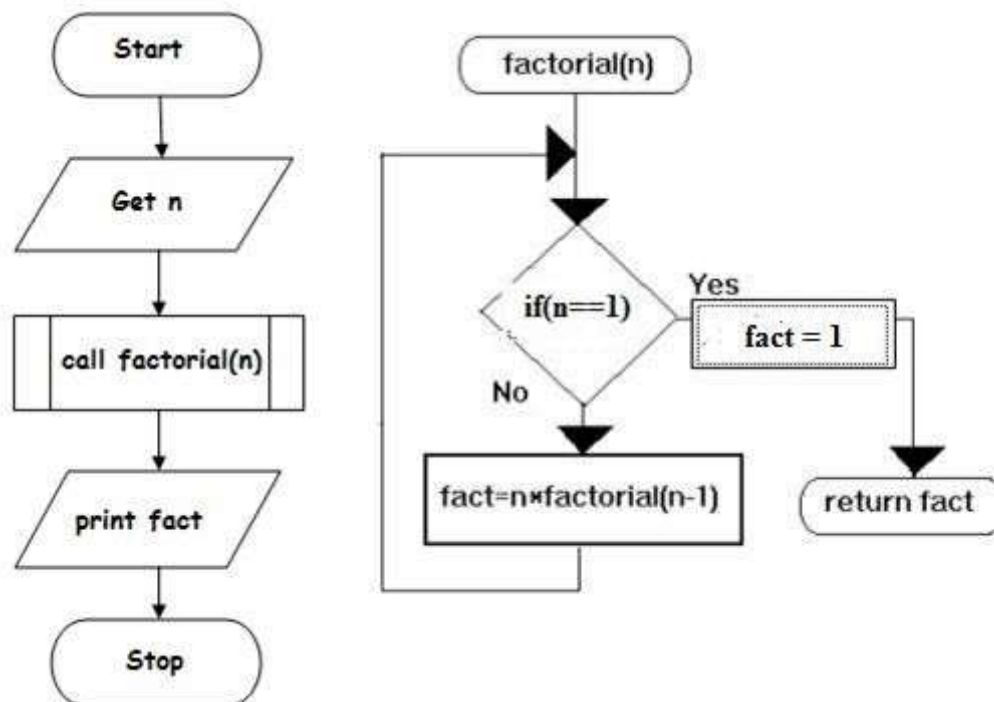
Algorithm for factorial of n numbers using recursion:

Main function:

Step1: Start
Step2: Get n
Step3: call factorial(n)
Step4: print fact
Step5: Stop

Sub function factorial(n):

Step1: if(n==1) then fact=1 return fact
Step2: else fact=n*factorial(n-1) and return fact



Pseudo code for factorial using recursion:

Main function:

BEGIN
GET n
CALL factorial(n)
PRINT fact
BIN

Sub function factorial(n):

```

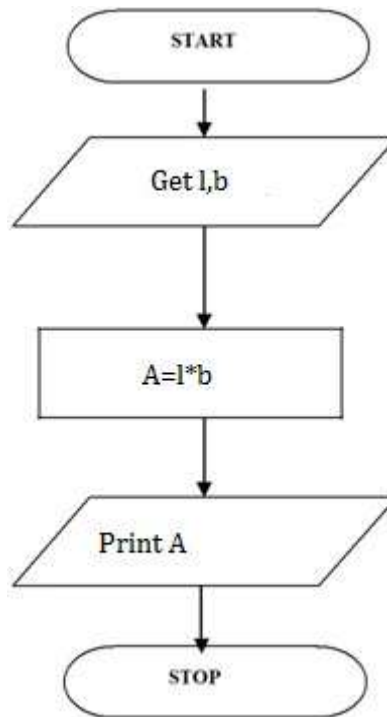
IF(n==1) THEN
    fact=1
    RETURN fact
ELSE
    RETURN fact=n*factorial(n-1)

```

More examples:

Write an algorithm to find area of a rectangle

Step 1: Start
Step 2: get l,b values
Step 3: Calculate $A=l*b$
Step 4: Display A
Step 5: Stop



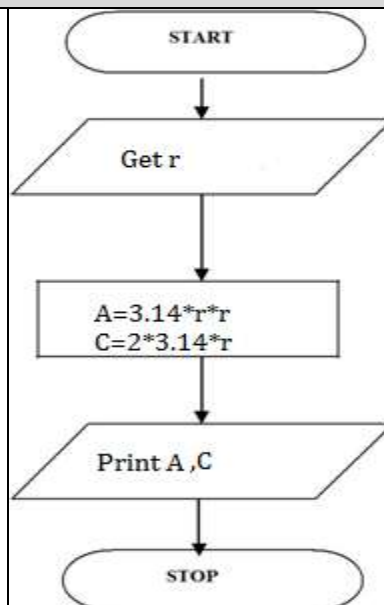
```

BEGIN
READ l,b
CALCULATE A=l*b
DISPLAY A
END

```

Write an algorithm for Calculating area and circumference of circle

Step 1: Start
Step 2: get r value
Step 3: Calculate $A=3.14*r*r$
Step 4: Calculate $C=2*3.14*r$
Step 5: Display A,C
Step 6: Stop



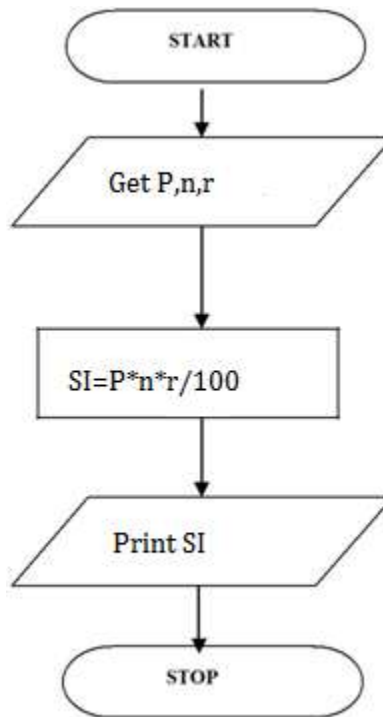
```

BEGIN
READ r
CALCULATE A and C
A=3.14*r*r
C=2*3.14*r
DISPLAY A
END

```

Write an algorithm for Calculating simple interest

Step 1: Start
Step 2: get P, n, r value
Step3: Calculate
 $SI = (p * n * r) / 100$
Step 4: Display S
Step 5: Stop

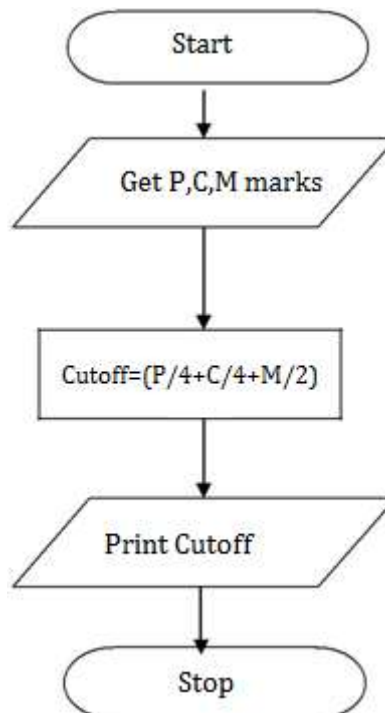


```

BEGIN
READ P, n, r
CALCULATE S
SI=(p*n*r)/100
DISPLAY SI
END
    
```

Write an algorithm for Calculating engineering cutoff

Step 1: Start
Step2: get P,C,M value
Step3: calculate
 $Cutoff = (P/4 + C/4 + M/2)$
Step 4: Display Cutoff
Step 5: Stop



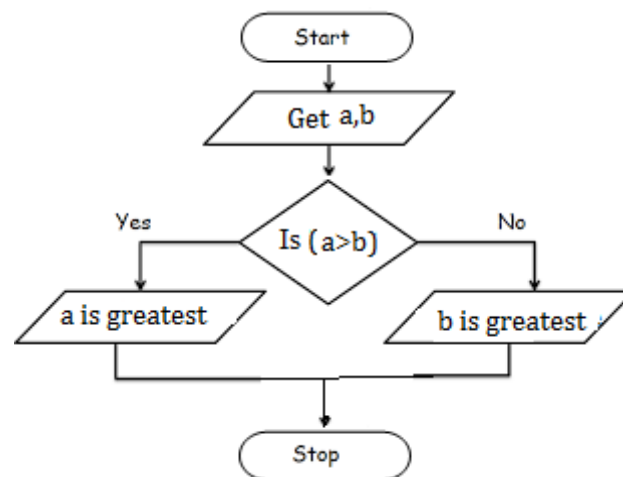
```

BEGIN
READ P,C,M
CALCULATE
Cutoff= (P/4+C/4+M/2)
DISPLAY Cutoff
END
    
```

To check greatest of two numbers

Step 1: Start
Step 2: get a,b value
Step 3: check if(a>b) print a is greater
Step 4: else b is greater
Step 5: Stop

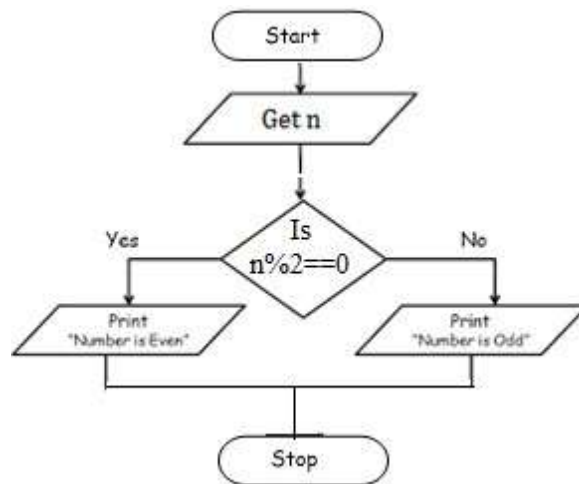
```
BEGIN
READ a,b
IF (a>b) THEN
DISPLAY a is greater
ELSE
DISPLAY b is greater
END IF
END
```



To check leap year or not

Step 1: Start
Step 2: get y
Step 3: if(y%4==0) print leap year
Step 4: else print not leap year
Step 5: Stop

```
BEGIN
READ y
IF (y%4==0) THEN
DISPLAY leap year
ELSE
DISPLAY not leap year
END IF
END
```



To check positive or negative number

Step 1: Start

Step 2: get num

Step 3: check if(num>0) print a is positive

Step 4: else num is negative

Step 5: Stop

BEGIN

READ num

IF (num>0) THEN

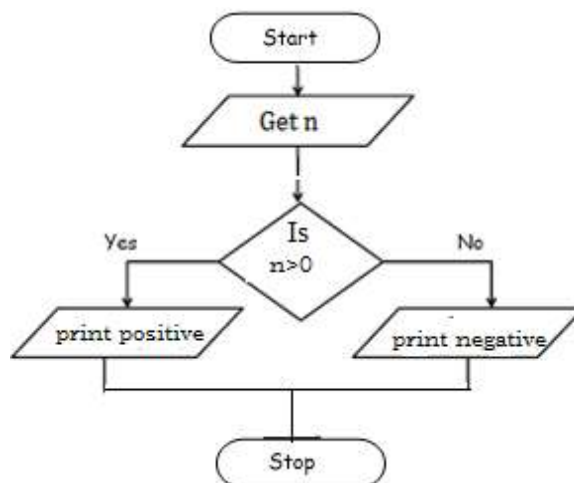
DISPLAY num is positive

ELSE

DISPLAY num is negative

END IF

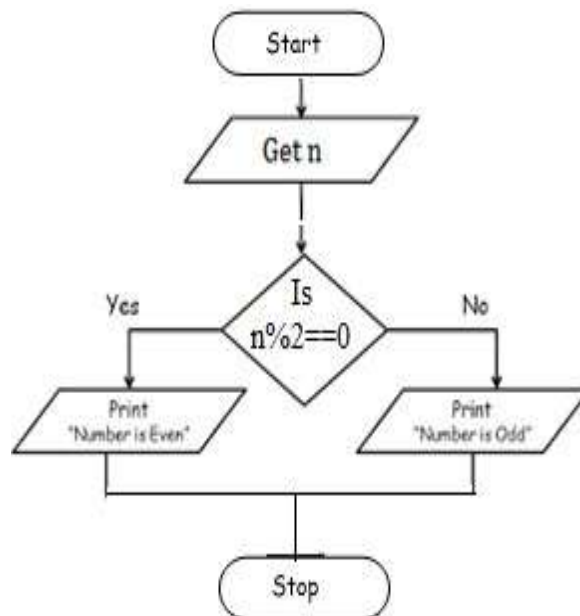
END



To check odd or even number

Step 1: Start
Step 2: get num
Step 3: check if($\text{num} \% 2 == 0$) print num is even
Step 4: else num is odd
Step 5: Stop

```
BEGIN
READ num
IF ( $\text{num} \% 2 == 0$ ) THEN
DISPLAY num is even
ELSE
DISPLAY num is odd
END IF
END
```



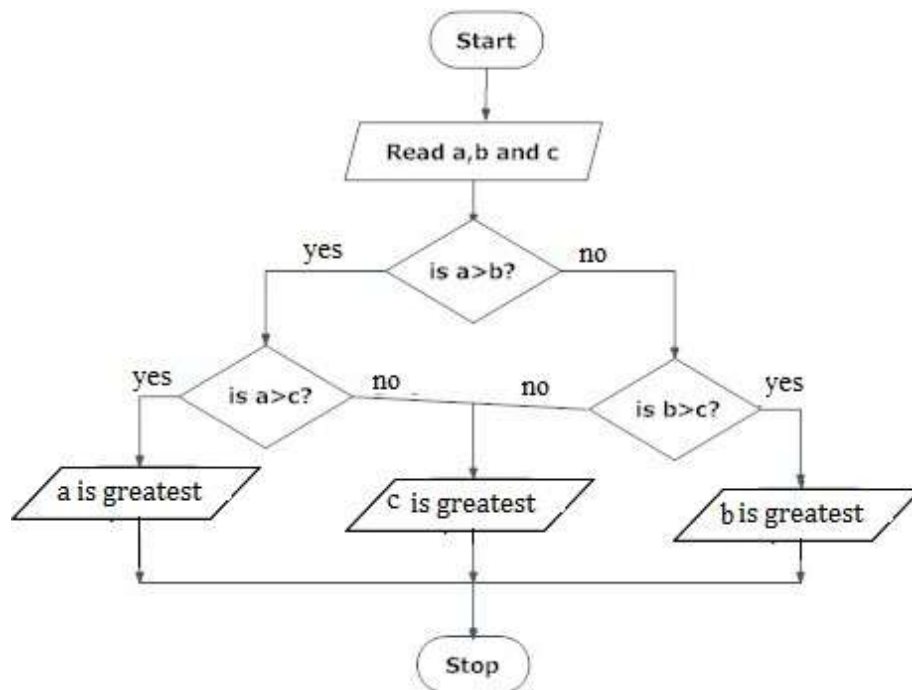
To check greatest of three numbers

Step1: Start
Step2: Get A, B, C
Step3: if($A > B$) goto Step4 else goto step5
Step4: If($A > C$) print A else print C
Step5: If($B > C$) print B else print C
Step6: Stop


```

BEGIN
READ a, b, c
IF (a>b) THEN
    IF(a>c) THEN
        DISPLAY a is greater
    ELSE
        DISPLAY c is greater
    END IF
ELSE
    IF(b>c) THEN
        DISPLAY b is greater
    ELSE
        DISPLAY c is greater
    END IF
END IF
END

```



Write an algorithm to check whether given number is +ve, -ve or zero.

Step 1: Start

Step 2: Get n value.

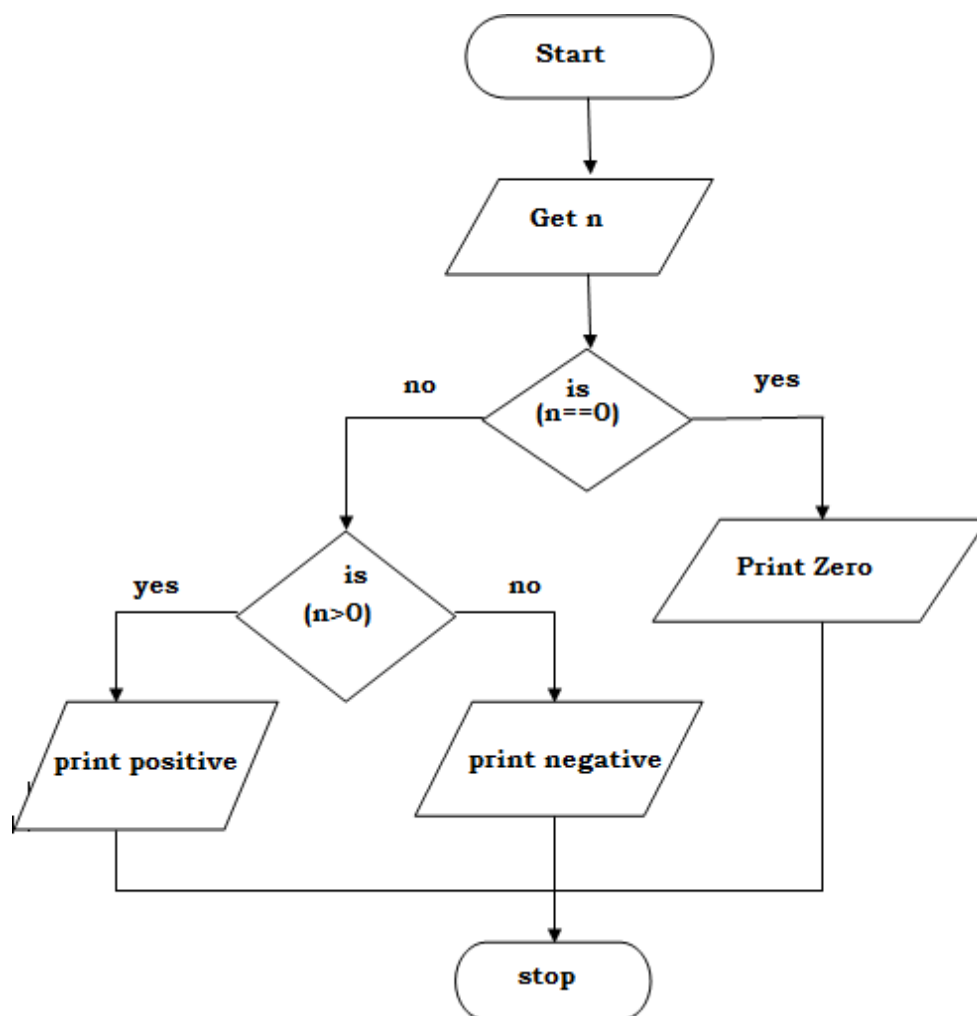
Step 3: if (n == 0) print "Given number is Zero" Else goto step4

Step 4: if (n > 0) then Print "Given number is +ve"

Step 5: else Print "Given number is -ve"

Step 6: Stop

```
BEGIN
GET n
IF(n==0) THEN
    DISPLAY " n is zero"
ELSE
    IF(n>0) THEN
        DISPLAY "n is positive"
    ELSE
        DISPLAY "n is positive"
    END IF
END IF
END
```



Write an algorithm to print all natural numbers up to n

Step 1: Start

Step 2: get n value.

Step 3: initialize $i=1$

Step 4: if ($i \leq n$) go to step 5 else go to step 8

Step 5: Print i value

step 6 : increment i value by 1

Step 7: go to step 4

Step 8: Stop

BEGIN

GET n

INITIALIZE $i=1$

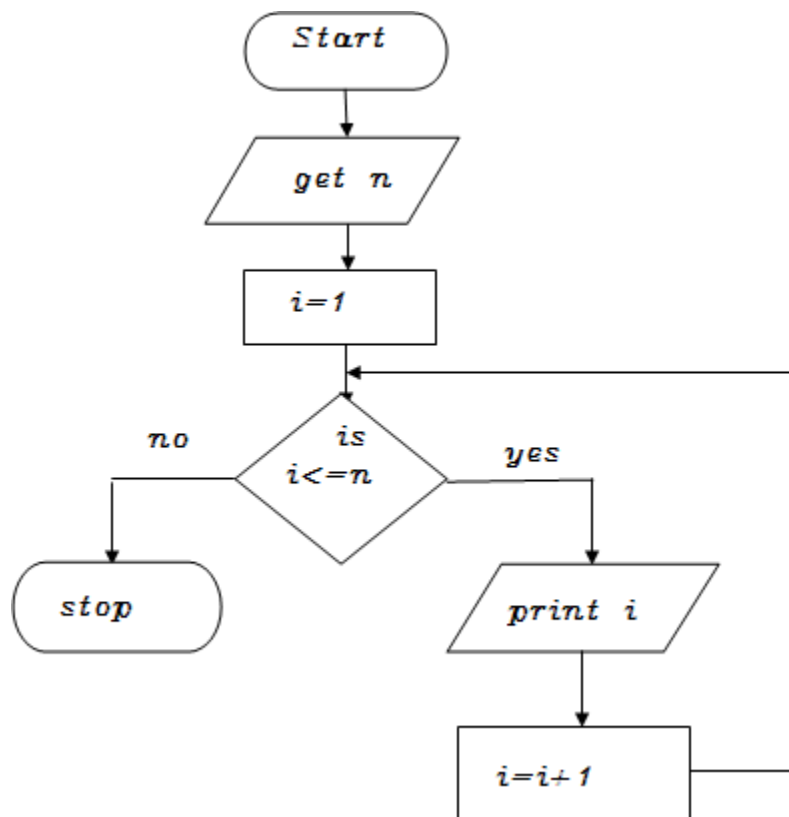
WHILE($i \leq n$) DO

 PRINT i

$i=i+1$

ENDWHILE

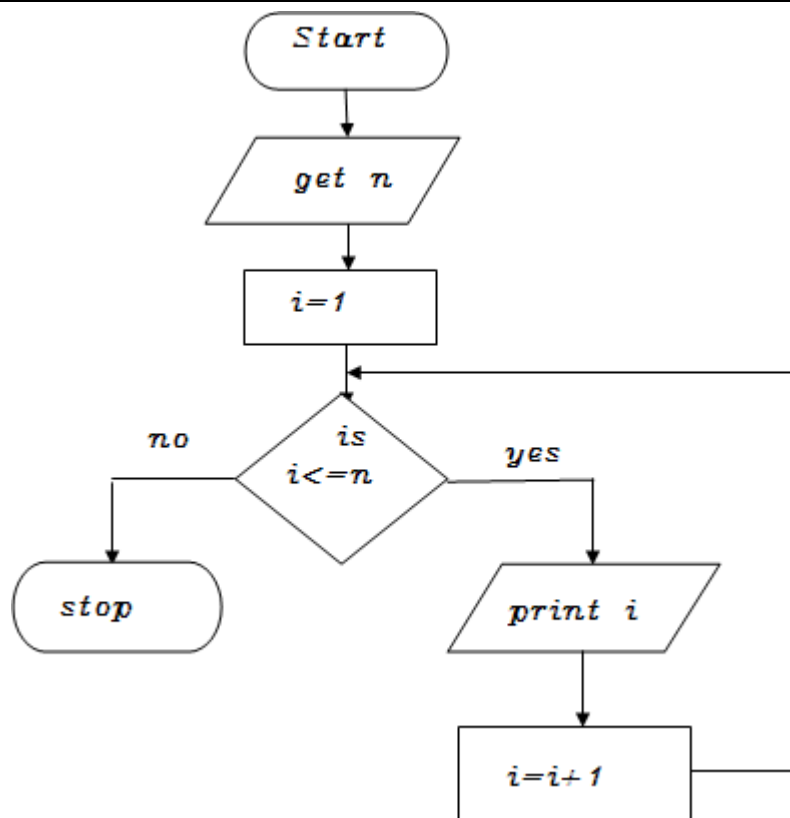
END



Write an algorithm to print n odd numbers

Step 1: start
step 2: get n value
step 3: set initial value $i=1$
step 4: check if($i \leq n$) goto step 5 else goto step 8
step 5: print i value
step 6: increment i value by 2
step 7: goto step 4
step 8: stop

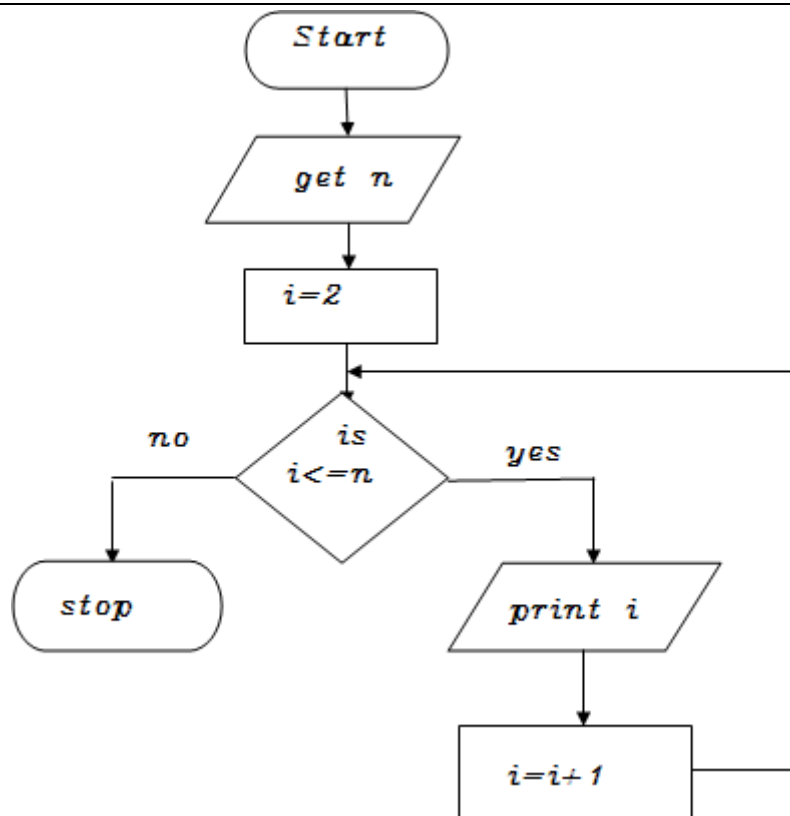
```
BEGIN
GET n
INITIALIZE i=1
WHILE(i<=n) DO
    PRINT i
    i=i+2
ENDWHILE
END
```



Write an algorithm to print n even numbers

Step 1: start
step 2: get n value
step 3: set initial value $i=2$
step 4: check if($i \leq n$) goto step 5 else goto step 8
step 5: print i value
step 6: increment i value by 2
step 7: goto step 4
step 8: stop

```
BEGIN
GET n
INITIALIZE i=2
WHILE(i<=n) DO
    PRINT i
    i=i+2
ENDWHILE
END
```



Write an algorithm to print squares of a number

Step 1: start

step 2: get n value

step 3: set initial value $i=1$

step 4: check i value if($i \leq n$) goto step 5 else goto step8

step 5: print $i*i$ value

step 6: increment i value by 1

step 7: goto step 4

step 8: stop

BEGIN

GET n

INITIALIZE $i=1$

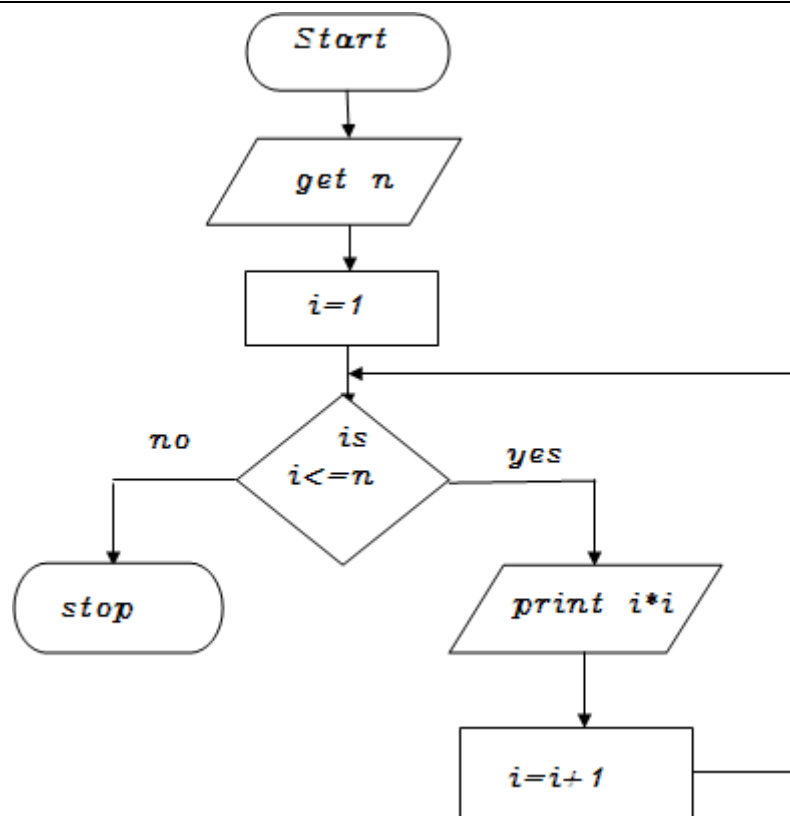
WHILE($i \leq n$) DO

 PRINT $i*i$

$i=i+1$

ENDWHILE

END



Write an algorithm to print to print cubes of a number

Step 1: start

step 2: get n value

step 3: set initial value $i=1$

step 4: check i value if($i \leq n$) goto step 5 else goto step8

step 5: print $i*i*i$ value

step 6: increment i value by 1

step 7: goto step 4

step 8: stop

BEGIN

GET n

INITIALIZE $i=1$

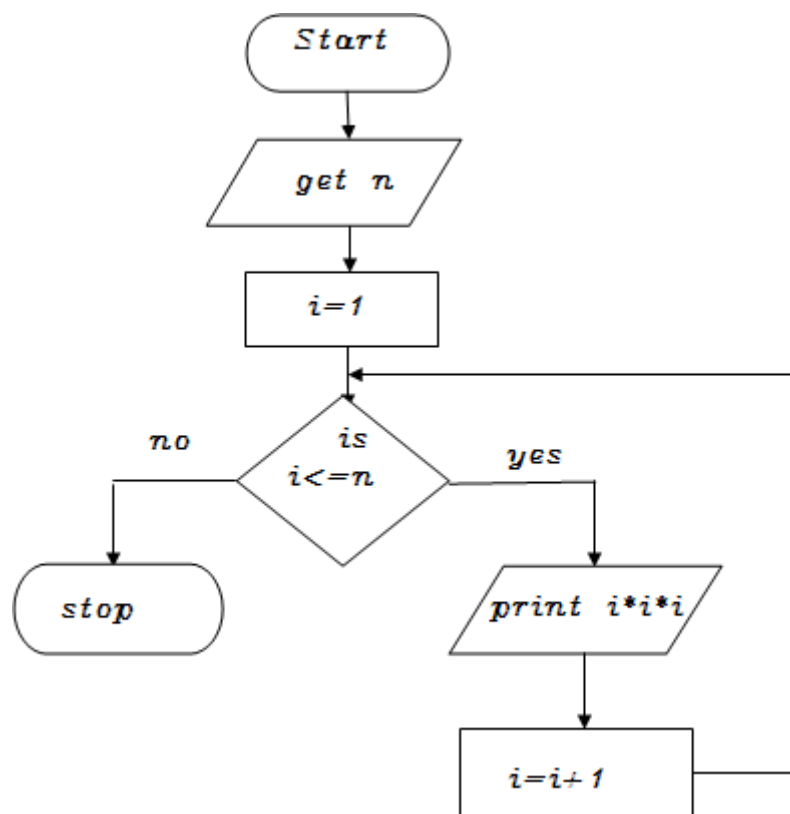
WHILE($i \leq n$) DO

 PRINT $i*i*i$

$i=i+1$

ENDWHILE

END



Write an algorithm to find sum of a given number

Step 1: start

step 2: get n value

step 3: set initial value $i=1$, $sum=0$

Step 4: check i value if($i \leq n$) goto step 5 else goto step8

step 5: calculate $sum=sum+i$

step 6: increment i value by 1

step 7: goto step 4

step 8: print sum value

step 9: stop

BEGIN

GET n

INITIALIZE $i=1, sum=0$

WHILE($i \leq n$) DO

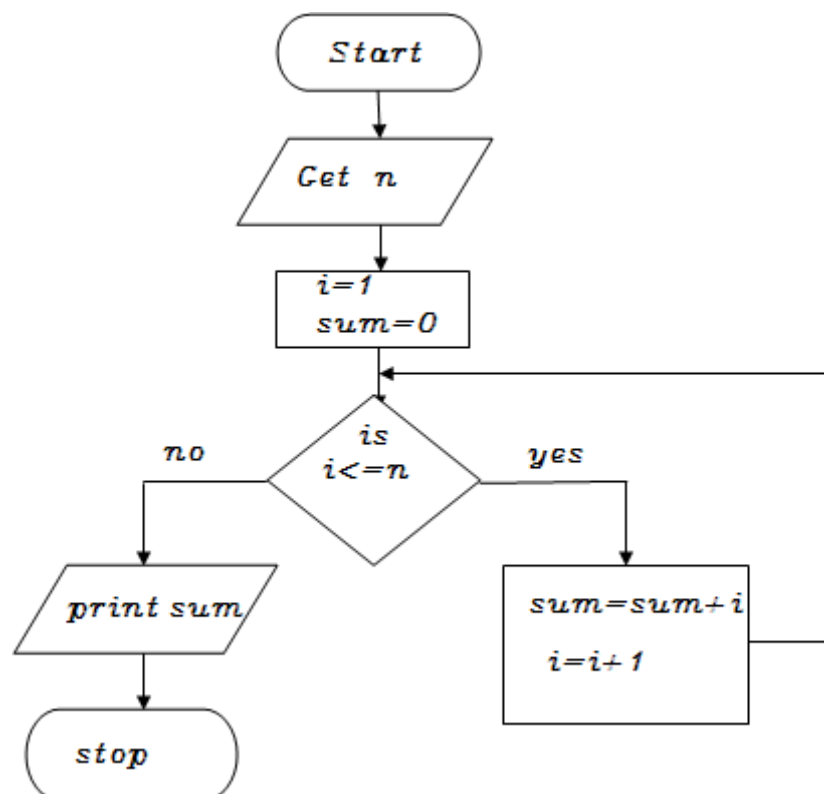
$sum=sum+i$

$i=i+1$

ENDWHILE

PRINT sum

END



Write an algorithm to find factorial of a given number

Step 1: start

step 2: get n value

step 3: set initial value $i=1$, $fact=1$

Step 4: check i value if($i \leq n$) goto step 5 else goto step8

step 5: calculate $fact=fact*i$

step 6: increment i value by 1

step 7: goto step 4

step 8: print fact value

step 9: stop

BEGIN

GET n

INITIALIZE $i=1, fact=1$

WHILE($i \leq n$) DO

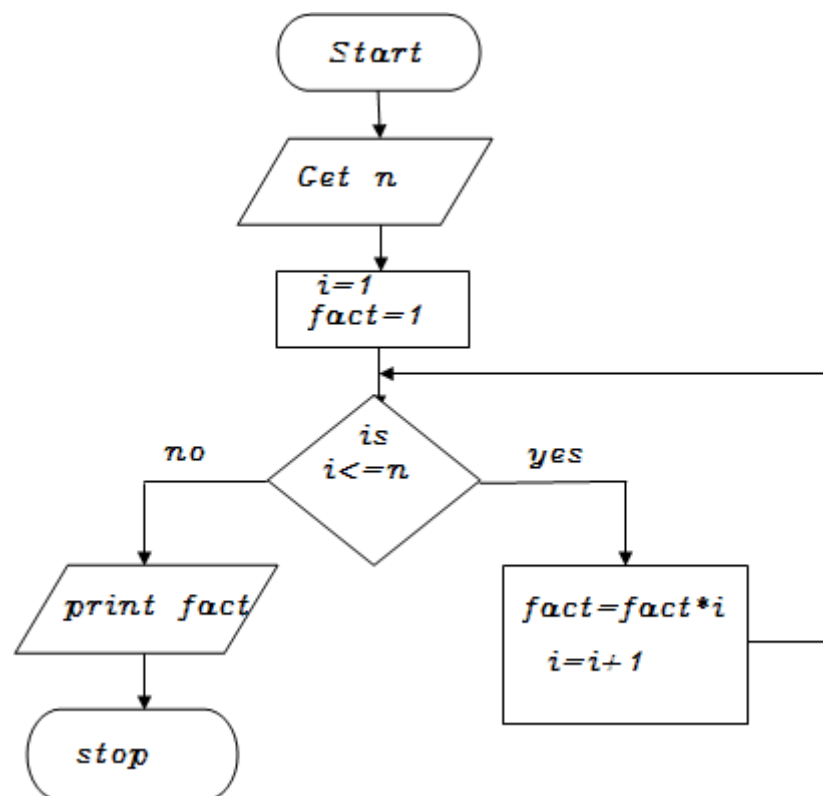
$fact=fact*i$

$i=i+1$

ENDWHILE

PRINT fact

END



Basic python programs:

Addition of two numbers	Output
<pre>a=eval(input("enter first no")) b=eval(input("enter second no")) c=a+b print("the sum is ",c)</pre>	<pre>enter first no 5 enter second no 6 the sum is 11</pre>
Area of rectangle	Output
<pre>l=eval(input("enter the length of rectangle")) b=eval(input("enter the breath of rectangle")) a=l*b print(a)</pre>	<pre>enter the length of rectangle 5 enter the breath of rectangle 6 30</pre>
Area & circumference of circle	output
<pre>r=eval(input("enter the radius of circle")) a=3.14*r*r c=2*3.14*r print("the area of circle",a) print("the circumference of circle",c)</pre>	<pre>enter the radius of circle4 the area of circle 50.24 the circumference of circle 25.12</pre>
Calculate simple interest	Output
<pre>p=eval(input("enter principle amount")) n=eval(input("enter no of years")) r=eval(input("enter rate of interest")) si=p*n*r/100 print("simple interest is",si)</pre>	<pre>enter principle amount 5000 enter no of years 4 enter rate of interest6 simple interest is 1200.0</pre>
Calculate engineering cutoff	Output
<pre>p=eval(input("enter physics marks")) c=eval(input("enter chemistry marks")) m=eval(input("enter maths marks")) cutoff=(p/4+c/4+m/2) print("cutoff =",cutoff)</pre>	<pre>enter physics marks 100 enter chemistry marks 99 enter maths marks 96 cutoff = 97.75</pre>
Check voting eligibility	output
<pre>age=eval(input("enter ur age")) If(age>=18): print("eligible for voting") else: print("not eligible for voting")</pre>	<pre>Enter ur age 19 Eligible for voting</pre>

Find greatest of three numbers	output
<pre> a=eval(input("enter the value of a")) b=eval(input("enter the value of b")) c=eval(input("enter the value of c")) if(a>b): if(a>c): print("the greatest no is",a) else: print("the greatest no is",c) else: if(b>c): print("the greatest no is",b) else: print("the greatest no is",c) </pre>	enter the value of a 9 enter the value of a 1 enter the value of a 8 the greatest no is 9
Programs on for loop	
Print n natural numbers	Output
<pre> for i in range(1,5,1): print(i) </pre>	1 2 3 4
Print n odd numbers	Output
<pre> for i in range(1,10,2): print(i) </pre>	1 3 5 7 9
Print n even numbers	Output
<pre> for i in range(2,10,2): print(i) </pre>	2 4 6 8
Print squares of numbers	Output
<pre> for i in range(1,5,1): print(i*i) </pre>	1 4 9 16
Print squares of numbers	Output
<pre> for i in range(1,5,1): print(i*i*i) </pre>	1 8 27 64
Programs on while loop	

Print n natural numbers	Output
<pre>i=1 while(i<=5): print(i) i=i+1</pre>	<pre>1 2 3 4 5</pre>
Print n odd numbers	Output
<pre>i=2 while(i<=10): print(i) i=i+2</pre>	<pre>2 4 6 8 10</pre>
Print n even numbers	Output
<pre>i=1 while(i<=10): print(i) i=i+2</pre>	<pre>1 3 5 7 9</pre>
Print n squares of numbers	Output
<pre>i=1 while(i<=5): print(i*i) i=i+1</pre>	<pre>1 4 9 16 25</pre>
Print n cubes numbers	Output
<pre>i=1 while(i<=3): print(i*i*i) i=i+1</pre>	<pre>1 8 27</pre>
find sum of n numbers	Output
<pre>i=1 sum=0 while(i<=10): sum=sum+i i=i+1 print(sum)</pre>	<pre>55</pre>

factorial of n numbers/product of n numbers	Output
<pre>i=1 product=1 while(i<=10): product=product*i i=i+1 print(product)</pre>	3628800
sum of n numbers	Output
<pre>def add(): a=eval(input("enter a value")) b=eval(input("enter b value")) c=a+b print("the sum is",c) add()</pre>	enter a value 6 enter b value 4 the sum is 10
area of rectangle using function	Output
<pre>def area(): l=eval(input("enter the length of rectangle")) b=eval(input("enter the breath of rectangle")) a=l*b print("the area of rectangle is",a) area()</pre>	enter the length of rectangle 20 enter the breath of rectangle 5 the area of rectangle is 100
swap two values of variables	Output
<pre>def swap(): a=eval(input("enter a value")) b=eval(input("enter b value")) c=a a=b b=c print("a=",a,"b=",b) swap()</pre>	enter a value3 enter b value5 a= 5 b= 3

check the no divisible by 5 or not	Output
<pre>def div(): n=eval(input("enter n value")) if(n%5==0): print("the number is divisible by 5") else: print("the number not divisible by 5") div()</pre>	enter n value10 the number is divisible by 5
find reminder and quotient of given no	Output
<pre>def reminder(): a=eval(input("enter a")) b=eval(input("enter b")) R=a%b print("the reminder is",R) def quotient(): a=eval(input("enter a")) b=eval(input("enter b")) Q=a/b print("the reminder is",Q) reminder() quotient()</pre>	enter a 6 enter b 3 the reminder is 0 enter a 8 enter b 4 the reminder is 2.0
convert the temperature	Output
<pre>def ctof(): c=eval(input("enter temperature in centigrade")) f=(1.8*c)+32 print("the temperature in Fahrenheit is",f) def ftoc(): f=eval(input("enter temp in Fahrenheit")) c=(f-32)/1.8 print("the temperature in centigrade is",c) ctof() ftoc()</pre>	enter temperature in centigrade 37 the temperature in Fahrenheit is 98.6 enter temp in Fahrenheit 100 the temperature in centigrade is 37.77

program for basic calculator	Output
<pre>def add(): a=eval(input("enter a value")) b=eval(input("enter b value")) c=a+b print("the sum is",c) def sub(): a=eval(input("enter a value")) b=eval(input("enter b value")) c=a-b print("the diff is",c) def mul(): a=eval(input("enter a value")) b=eval(input("enter b value")) c=a*b print("the mul is",c) def div(): a=eval(input("enter a value")) b=eval(input("enter b value")) c=a/b print("the div is",c) add() sub() mul() div()</pre>	<pre>enter a value 10 enter b value 10 the sum is 20 enter a value 10 enter b value 10 the diff is 0 enter a value 10 enter b value 10 the mul is 100 enter a value 10 enter b value 10 the div is 1</pre>